Windows Server Infrastructure & Application Delivery Troubleshooting Report

Sample Work — Demonstration Environment

Project Overview

This case study documents troubleshooting and configuration work performed in an AWS-hosted Windows Server environment supporting Active Directory, Amazon FSx file storage, Parallels RAS application delivery, and MSIX application packages.

The work involved identifying and resolving issues across the Windows Server infrastructure, shared application storage, application packaging, and application runtime dependencies.

Environment

- **Cloud Platform:** Amazon Web Services (AWS)
- **Server Platform:** Windows Server 2022
- **Directory Services:** Active Directory
- **Storage:** Amazon FSx for Windows File Server
- **Application Delivery:** Parallels RAS
- **Application Packaging:** MSIX / VHDX
- **Applications Tested:** LibreOffice and other packaged applications
- **Client Environment:** Windows-based virtual desktop/application delivery environment

Issue 1 — Active Directory and File Storage Integration

Problem

The Windows Server environment needed to provide centralized identity services while allowing application packages and related files to be accessed from Amazon FSx.

The environment required proper communication between the Active Directory infrastructure, Windows Server systems, and the FSx SMB file share.

Investigation

The environment was examined from the Windows Server side to verify:

- Active Directory availability
- Server connectivity
- DNS resolution
- Domain membership
- Access to the FSx SMB share
- File and folder accessibility
- Access to application package files stored on FSx

The FSx share was successfully accessible from the Windows Server environment and could be mapped and browsed through Windows Explorer.

The application package files, including MSIX/VHDX files, were confirmed to be present on the FSx storage.

Findings

The investigation confirmed that the underlying Windows Server and FSx storage environment was operational and that the application package files were available from the shared storage location.

Additional troubleshooting was required around how Parallels RAS handled the shared storage path when working with MSIX application packages.

Resolution

The FSx storage was established as the shared repository for the application package files, allowing the Windows Server/RAS environment to access the packages from centralized storage.

The distinction between standard application publication and MSIX image handling in Parallels RAS was also identified during troubleshooting.

Issue 2 — MSIX/VHDX Application Deployment

Problem

MSIX/VHDX application packages were successfully stored on the FSx share and made available to the application delivery environment.

The applications appeared correctly in the Windows Start Menu, indicating that the VHDX was being attached and the application package was being registered.

However, launching the applications failed.

Investigation

The troubleshooting process included:

- Verifying the MSIX/VHDX files on the FSx share
- Confirming that the VHDX packages were being attached
- Confirming that the applications appeared in the Start Menu
- Reviewing Windows Event Viewer
- Reviewing application deployment and runtime errors
- Checking application dependencies
- Testing the packaged application directly on the Windows Server/VDI environment

The fact that the applications appeared in the Start Menu was an important diagnostic indicator.

It showed that the package was being successfully recognized and registered, shifting the investigation away from the initial package-storage and attachment process and toward application runtime dependencies.

Findings

Event Viewer identified a dependency involving **Microsoft.WindowsAppRuntime**.

The LibreOffice MSIX package declared Microsoft Windows App Runtime as a separate framework dependency. The required runtime version was not available on the target environment.

This meant that the MSIX package itself was not necessarily defective. The application was being deployed successfully, but the required runtime component needed to launch the application was missing.

Resolution

The troubleshooting process identified that **Microsoft.WindowsAppRuntime 1.4** needed to be installed on the target environment.

Rather than rebuilding the LibreOffice MSIX package, the required Windows App Runtime dependency was installed separately.

After the required runtime dependency was available, the application deployment process could proceed without rebuilding the application package solely to address the missing runtime.

Issue 3 — MSIX Package Path and Shared Storage

Problem

Another issue involved how Parallels RAS handled application package locations.

The FSx share was available as a mapped drive and the MSIX/VHDX files could be browsed through Windows.

However, Parallels RAS handled normal application publication paths differently from the MSIX Image path configuration.

Investigation

Testing showed that:

- The FSx SMB share was accessible through a mapped drive.
- VHDX files were present on the share.
- Standard RAS publication could work with UNC-based paths.
- The MSIX Image path selection behavior differed from normal application publication.
- The RAS interface could favor the mapped-drive representation when selecting MSIX image locations.

The troubleshooting therefore focused not only on whether the files were accessible, but also on how the application delivery platform expected the package repository to be referenced.

Findings

The issue was determined to involve the interaction between the shared FSx repository and the way Parallels RAS handled MSIX image paths.

This demonstrated the importance of distinguishing between:

File accessibility

and

Application delivery platform path requirements.

A file being accessible from Windows Explorer does not automatically mean that every application delivery component will handle that same path in the same way.

Resolution

The FSx repository and application package paths were tested through the RAS application delivery configuration, with attention to UNC and mapped-drive path handling.

The troubleshooting established the correct relationship between the shared application repository and the RAS application package configuration.

Troubleshooting Approach

The troubleshooting process followed a structured progression rather than immediately rebuilding components.

1. Verify the Infrastructure

The Windows Server environment, Active Directory, network connectivity, and FSx storage were checked first.

2. Verify File Availability

The MSIX/VHDX application files were confirmed to exist on the FSx share and were accessible from the Windows environment.

3. Verify Application Registration

The applications were checked in the Windows Start Menu to determine whether the MSIX/VHDX deployment process was completing successfully.

4. Review System Evidence

Windows Event Viewer was used to identify errors occurring when the applications were launched.

5. Identify Dependencies

The investigation determined that the application had a Microsoft.WindowsAppRuntime dependency that was not available on the target environment.

6. Correct the Dependency

Microsoft.WindowsAppRuntime 1.4 was installed on the target environment rather than unnecessarily rebuilding the application package.

7. Verify Application Delivery Configuration

The interaction between FSx storage, package paths, and Parallels RAS application package configuration was tested and reviewed.

Technologies Used

Infrastructure

- Amazon Web Services
- EC2
- Windows Server 2022
- Active Directory
- DNS
- Amazon FSx for Windows File Server

Application Delivery

- Parallels RAS
- MSIX
- VHDX
- Microsoft Windows App Runtime
- Application packaging and deployment

Troubleshooting Tools

- Windows Server Manager
- Event Viewer
- PowerShell
- Windows Explorer
- Active Directory Users and Computers
- Parallels RAS management tools

Result

The troubleshooting process successfully identified multiple points of failure across the infrastructure and application delivery workflow.

The investigation established that:

- The Windows Server environment was operational.
- Active Directory services were functioning within the environment.
- Amazon FSx was providing centralized SMB storage for application packages.
- MSIX/VHDX packages were successfully being accessed and registered.
- Application launch failures were traced to a missing Microsoft.WindowsAppRuntime dependency.
- Microsoft.WindowsAppRuntime 1.4 was identified as the required runtime dependency.
- The application package did not need to be unnecessarily rebuilt to address the runtime dependency.
- Parallels RAS MSIX image path behavior was separately identified and investigated.

Key Takeaway

The project required troubleshooting across multiple layers of the environment rather than relying on a single configuration change.

The investigation moved from **infrastructure → storage → application package → registration → event logs → runtime dependencies → application delivery configuration** to isolate the actual problems.

This approach helps reduce unnecessary changes and allows the underlying cause of a Windows Server or application delivery problem to be identified before implementing a solution.

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